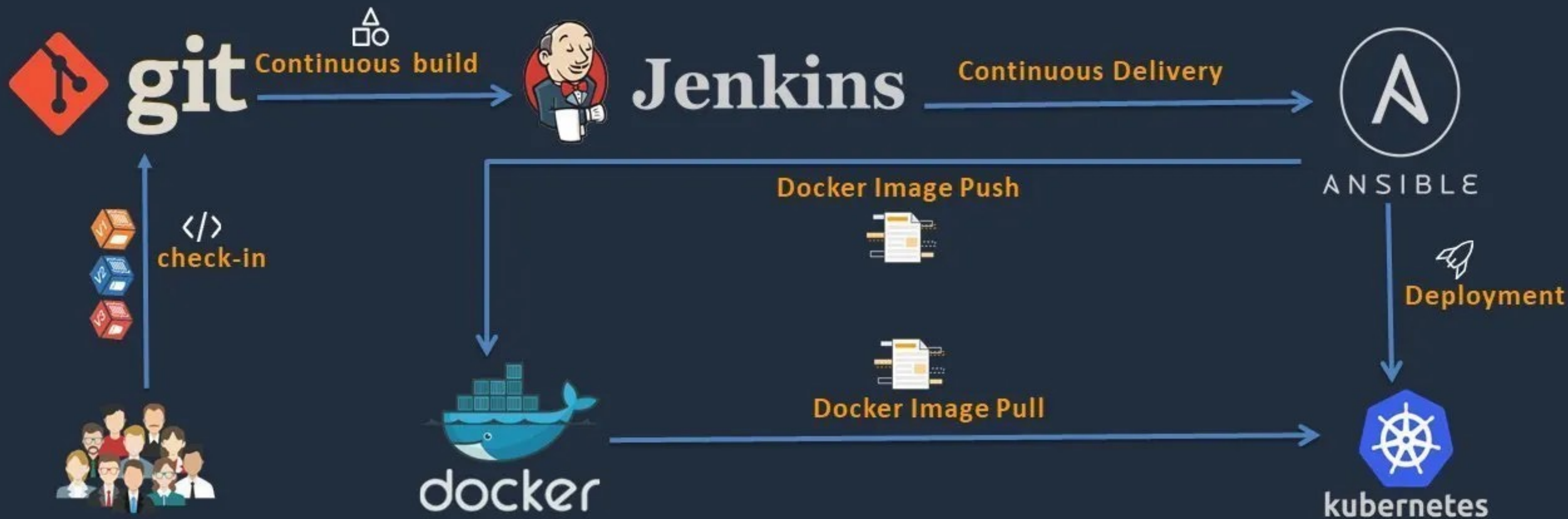


DevOps Project



```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: addressbook-deployment

spec:
  replicas: 1

  selector:
    matchLabels:
      app: addressbook

  template:
    metadata:
      labels:
        app: addressbook

    spec:
      containers:

        # ■ Main App (Tomcat)
        - name: addressbook
          image: tomcat:9-jdk17
          ports:
            - containerPort: 8080
```

```
      volumeMounts:
        - name: war-volume
          mountPath: /usr/local/tomcat/webapps/addressbook.war

      # ■ Node Exporter
      - name: node-exporter
        image: prom/node-exporter
        ports:
          - containerPort: 9100

      volumes:
        - name: war-volume
          hostPath:
            path: /home/ubuntu/target/addressbook.war # ✓ ABSOLU
            type: File
```



```
apiVersion: v1
kind: Service
metadata:
  name: addressbook-service
  labels:
    app: addressbook

spec:
  type: NodePort
  selector:
    app: addressbook

  ports:
    - name: http
      port: 8080
      targetPort: 8080
      nodePort: 30007

    - name: metrics
      port: 9100
      targetPort: 9100
```

```
apiVersion: monitoring.coreos.com/v1
kind: ServiceMonitor
metadata:
  name: addressbook-monitor
  labels:
    release: monitoring

spec:
  selector:
    matchLabels:
      app: addressbook

  namespaceSelector:
    matchNames:
      - default

  endpoints:
    - port: metrics
      path: /metrics
      interval: 5s
```

```
grafana:
  service:
    type: NodePort
    nodePort: 30009

  adminUser: admin
  adminPassword: "admin123"

  sidecar:
    dashboards:
      enabled: true

prometheus:
  service:
    type: NodePort
    nodePort: 30008
```

```
plugin: amazon.aws.aws_ec2

regions:
| - us-east-1

filters:
| "tag:Roles": "app-server"

hostnames:
| - public-dns-name

keyed_groups:
| - key: tags.Name
|   prefix: server

compose:
| ansible_host: public_ip_address
| ansible_user: "'ubuntu'"
```



```
---
- name: ContactBook's Full DevOps Setup (K3s + Helm + Monitoring + App)
  hosts: all
  become: yes

  environment:
    KUBECONFIG: /etc/rancher/k3s/k3s.yaml

  tasks:
    - name: Copy K8s files to server
      copy:
        src: /Users/jagdish/Documents/DevOpsClassCodes/k8s/
        dest: /home/ubuntu/k8s/

    - name: Copy Monitoring files
      copy:
        src: /Users/jagdish/Documents/DevOpsClassCodes/monitoring/
        dest: /home/ubuntu/monitoring/

    - name: Copy Output
      copy:
        src: ../target/addressbook.war
        dest: /home/ubuntu/target/
```


- **name:** Install K3s
shell: `curl -sfL https://get.k3s.io | sh -`
args:
 - creates:** `/usr/local/bin/k3s`
- **name:** Wait for node to be ready
shell: `kubectl get nodes`
register: `result`
retries: `10`
delay: `10`
until: `result.rc == 0`
- **name:** Install Helm
shell: `curl https://raw.githubusercontent.com/helm/helm/main/scripts/get-helm-3 | bash`
args:
 - creates:** `/usr/local/bin/helm`

- **name:** Add Prometheus Helm repo
shell: |
helm repo add prometheus-community <https://prometheus-community.github.io/helm-charts>
helm repo update
- **name:** Install Prometheus Stack
shell: |
helm upgrade --install monitoring prometheus-community/kube-prometheus-stack \
-n monitoring \
--create-namespace \
-f /home/ubuntu/monitoring/values.yml
- **name:** Apply Deployment
command: kubectl apply -f /home/ubuntu/k8s/deployment.yml
- **name:** Apply Service
command: kubectl apply -f /home/ubuntu/k8s/service.yml

- **name:** Apply ServiceMonitor
command: kubectl apply -f /home/ubuntu/monitoring/serviceMonitor.yml
- **name:** Inject dashboard to Grafana
shell: |
 kubectl create configmap grafana-dashboard \
 --from-file=dashboard.json=/home/ubuntu/monitoring/dashboard.json \
 -n monitoring \
 --dry-run=client -o yaml | kubectl apply -f -
- **name:** Label configmap
shell: |
 kubectl label configmap grafana-dashboard \
 grafana_dashboard=1 \
 -n monitoring \
 --overwrite

```
pipeline{
  agent any

  tools{
    maven 'MyMaven'
    terraform 'Terraform'
    ansible 'ansible'
  }

  stages{
    stage('Checkout Code') {
      steps {
        git 'https://github.com/Jagdish-Patel1824/DevOpsClassCodes.git'
      }
    }

    stage('Build') {
      steps{
        sh 'mvn clean package'
      }
    }

    stage('TF Init') {
      steps{
        sh 'cd terraform && terraform init'
      }
    }
  }
}
```

```
stage('TF Apply') {  
    steps{  
        sh 'cd terraform && terraform apply -auto-approve'  
    }  
}
```

```
stage('Ansible Deploy') {  
    steps {  
        withCredentials([file(credentialsId: 'ec2-key', variable: 'ANSIBLE_PRIVATE_KEY_FILE')])  
        {  
            sh '''  
                export ANSIBLE_HOST_KEY_CHECKING=False  
                ansible-playbook -i ansible/inventory.aws_ec2.yml ansible/playbook.yml  
            '''  
        }  
    }  
}
```

```
}
```

```
}
```


Thank You